

ASP3231 Assignment 3

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Question 1

1a.

Wien's Displacement Law relates the wavelength at which a blackbody is most radiant to the inverse of its temperature. It is defined under the assumption that the object is a perfectly radiating blackbody, which is a typical assumption for analysing most stars. It follows that

$$\lambda_{\max} = \frac{2.898 \cdot 10^{-3} \text{m} \cdot \text{K}}{T}. \quad (1)$$

On the $t=1.43$ days graph in Figure 1, the wavelength that is emitted at the highest flux (intensity) is approximately $1.2 \cdot 10^3$ angstroms, or $1.2 \cdot 10^{-7}$ metres. Hence, the temperature of the object 1.43 days after the explosion is $\approx 2.415 \cdot 10^4$ K.

1b.

Wien's Displacement Law essentially tells us that an increase in the wavelength (lengthening) which corresponds to the brightest flux leads to a decrease in temperature. We can see that over time, the maximum wavelength is getting longer while the maximum intensity decreases, which means the temperature is decreasing over time. Using the same approximation as in Q1a, the temperature between the first and last epoch has decreased by approximately $1.84 \cdot 10^4$ K, where the temperature after 24.01 days is ≈ 5796 K.

1c.

We start to observe a significant deviation from the blackbody fit around the 14.94-day epoch. Beyond this there is a very significant deviation. This implies that the blackbody fit is no longer a reliable tool to estimate the temperature of the body.

In a more physical sense, the breaking down of the blackbody assumption implies that the supernova is losing opacity as it expands and cools. We start to lose the continuum emission of a perfect blackbody and see the individual emission lines start to dominate as the supernova becomes optically thin.

1d.

The flux at 10^4 angstroms of the object over time actually remains fairly constant. We can see that the last few data points, and even the blackbody fit, for the most part remain close to the 10^{-17} flux unit mark. The only noticeable shift is a slight increase at first followed by a slight decrease later.

For a blackbody, flux is directly proportional to temperature and the solid angle. Assuming the flux remains mostly constant, and the temperature is decreasing, the solid angle must be increasing over time. This physically represents the expansion of the supernova. Finding an exact size would require knowledge of the distance to the supernova.

1e.

Based on the previous estimates and given values we have a temperature of 24150 K, an f_λ of 10^{-9} erg $\text{s}^{-1} \text{cm}^{-3}$, a distance of $9.2 \cdot 10^{26}$ cm, and a wavelength of 10^{-4} cm (all in CGS units). The solid angle is calculated by

$$\Omega = \frac{\pi R^2}{d^2}. \quad (2)$$

We can then use this value where

$$f_\lambda = B_\lambda \Omega. \quad (3)$$

To get B_λ using our flux f_λ , we want to convert the provided B_ν formula to be per unit wavelength. This is given as

$$B_\lambda = B_\nu \frac{c}{\lambda^2} = \frac{2ckT}{\lambda^4}. \quad (4)$$

Alright, working backwards we can start with finding B_λ using Equation 4. c and k are provided constants, and we have the temperature and wavelength. Again, all values are in CGS. We have

$$B_\lambda = 1.998 \cdot 10^{15} \text{ erg s}^{-1} \text{ cm}^{-3} \text{ sr}^{-1}.$$

Putting B_λ and the approximate f_λ into Equation 3 will get us the solid angle

$$\Omega = 5.004 \cdot 10^{-25} \text{ sr}.$$

And lastly Equation 2 will give us the radius of the supernova, which comes out to

$$R = 3.67 \cdot 10^{14} \text{ cm} \approx 5276 \text{ solar radii}.$$

Question 2

2a.

The approximate value of f_λ at 3 microns is $0.8 \cdot 10^{-21}$ erg $\text{s}^{-1} \text{cm}^{-2} \text{\AA}^{-1}$. This is equivalent to $0.8 \cdot 10^{-20}$ W $\text{m}^{-2} \mu\text{m}^{-1}$. The energy of a single photon at $\lambda = 3$ microns is given by

$$E_{\text{photon}} = \frac{hc}{\lambda} = 6.62 \cdot 10^{-20} \text{ J}.$$

We can divide the flux by the energy of a single photon to obtain

$$\text{Flux Density} = \frac{f_\lambda}{E_{\text{photon}}} \approx 0.12 \text{ photons s}^{-1} \text{ m}^{-2} \mu\text{m}^{-1}.$$

2b.

It is given that the primary mirror diameter of JWST is 6.5 metres. Hence, the circular collecting area $A = \pi(\frac{d}{2})^2$ is approximately 33.18 m^2 . The counts in units of photons per pixel would be the number of source photons per second per square metre per micron (Q2a) multiplied by the collecting area, the exposure time, and the linear dispersion (microns per pixel). This works out as

$$\text{Counts} = 0.12 \cdot 33.18 \cdot 33612 \cdot 0.014 \approx 1874 \text{ photons/pixel}.$$

2c.

The ideal signal to noise, if dominated by Poisson noise, is given to be

$$\text{SNR}_{\text{max}} = \frac{N}{\sqrt{N}} = \sqrt{1874} \approx 43.$$

Looking at the figure, the observed noise spans roughly $0.4 \cdot 10^{-21}$ around a value of $0.8 \cdot 10^{-21}$, resulting in a signal to noise ratio of 2.

2d.

The observed signal to noise ratio was far lower than the idealised case. In reality, there is much more background radiation coming from the sky as well as thermal energy from the telescope, which all produce more noise. Another key reason is the read noise, which is introduced when counting the charge per pixel. Another physical limitation is the small slit size on the NIRSPEC which means that lots of the galaxy light won't be detected. All these reasons contribute to a much smaller signal to noise ratio.